

Valuing Minnesota Family Forests: Economic and Ecological Contributions

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Abstract

This report assesses the monetary and non-monetary value of Minnesota family forests, with particular attention to the ecosystem services they provide. - Should this be internal or external facing? - Treat this as developing a tool: either a framework or an application (ambitious) for policy recommendation decisions

Keywords: Family Forests; Ecosystem Services; Forest Valuation; Minnesota

1 Introduction

Describe the importance of Minnesota family forests, the decision context for this assessment, and the research questions addressed by the report. - Place within the strategic plan of MFRC. - Is there a broader (national) context to place this within.

2 The State of North America's Forests

Review the relevant literature, regional context, and ecosystem-service concepts that guide the analysis.

- Description of NA, national, and MN forests - Narrowing to family forests

2.1 Minnesota Family Forests

3 Methodology to Value Minnesota Family Forests

3.1 Study Scope

Define the study area, forest-owner population, ecosystem services, and time period included in the assessment.

3.2 Forest Ecosystems Assessed

- Board overview of ecosystem services - Specific ecosystem services evaluated - within the context of Gross-ecosystem product (GEP) value and natural capital accounting - This is not a strict accounting approach, but will attempt to be congruent with SEEA-EA accounting framework.

3.2.1 Ecosystem 1:

Data. Describe the datasets, their provenance, preparation, and any important limitations.

Valuation Approach. Explain the monetary and non-monetary valuation methods, assumptions, and analytical steps.

Estimated Value Present the estimated monetary and non-monetary values for the ecosystem service, including any relevant tables, figures, or summary statistics. Mainly supply, use, and monetary value.

3.2.2 Ecosystem 1: Timber Provision

Data. Smoothed (Krigged) map of forest characteristics across MN. FIA data on timber characteristics.

Valuation Approach. Lumber production given tree dimensions across forest. Use current market prices. Could potentially weight by market access (road network to nearest mills or major markets)

Estimated Value

3.2.3 Ecosystem 2: Carbon Sequestration and Storage

Data. Smoothed (Krigged) map of forest characteristics across MN. FIA carbon storage measure Carbon sequestration formula.

Valuation Approach. Social cost of carbon approach.

Estimated Value

3.2.4 Ecosystem 3: Habitat Support

Data. Bird populations. Species range maps, species abundance data. Biodiversity maps (often static estimates)

Valuation Approach. Either travel cost hedonic modeling. Or imputed value from literature review meta analysis.

Estimated Value

3.2.5 Ecosystem 4: Nature Recreation

Data. Hunting and fishing participation data. Visitor counts at recreational sites. Surveys on recreational preferences and expenditures.

Valuation Approach. Impute value from meta analysis of contingent valuation surveys. Trick will be to measure values from tree support of this recreation (e.g., variation in hunting value by tree characteristics) – simple OLS relationship.

Estimated Value

4 What is the Value of Minnesota Family Forests?

- TBD - General breakdown of value by groups (which ecosystem services dominate). Comparison of monetary and non-monetary values across ecosystem services. Do market forces drive the effect? - Want a visual comparison of timber production (bar) against stacked bar of all other ecosystem services. - Time trends. Is there anything to be stated about changes in forest type, ownership patterns, or ecosystem service provision over time? - Heterogeneity in forest types and acreage (size)

5 How Does Forest Value Change under a Warming Climate?

- Projections of climate change impacts on LULC - Implications for ecosystem service provision and forest value - What do we expect for transfer across ownership type (family forest to public or deforestation)? - What is the implication with agricultural expansion, prairie encroachment, increased global competition driving down timber demand (i.e., Eucalyptus spp.)

6 Policy Implications

- Given a fixed MN budget, what policies might we recommend? - How can policy support the sustainable management and conservation of family forests? - Are there trade-offs between different policy options in terms of ecosystem service provision and forest value?

7 Conclusion

Include limitations here. Summarize the principal findings and identify the most important next steps for research, data collection, or decision-making.

References

A Supplementary Appendix

Place detailed methods, assumptions, supporting tables, and additional figures here.

B Unassessed Ecosystem Services

List and describe ecosystem services that were not assessed in this study, explaining why they were excluded and any potential implications for the overall valuation of Minnesota family forests. Does this lead to underestimation of the total value of Minnesota family forests? - What are the gaps in the literature in what methodologies can be used or what data can be used? - Limitations to measurement.

C Methodological and Data Limitations

Here I want to document the limitations related to the methodology, data sources, assumptions, and any uncertainties that may affect the valuation of Minnesota family forests.

D Acknowledgments

Conflicts of Interest. The authors declare no conflicts of interest.

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